

Kritika Pandit



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Education

Carleton College

Expected June 2026

Bachelor of Arts in Computer Science (GPA: 3.71 / 4.00)

Northfield, Minnesota

- **Relevant Coursework:** Data Structures & Algorithms, Software Design, Systems, Linear Algebra, Discrete Math, Calculus Series, Data Science, Theory of Computing, Mobile App Development, Computer Security, Probability

Experience

Academic Technology, Carleton College

April 15 - September 11

DevSquad Junior Developer

Northfield, Minnesota

- Replaced Carleton's GPT-3.5 ITS helpdesk chatbot with an open-source Llama3 model, improving support response times by 40% and ensuring data security through PrivateGPT.
- Enhanced model performance and efficiency by implementing GPU acceleration, integrating the pgpt Python SDK to create a robust real-time server with streaming capabilities, and enabling multi-document injection to support over 1,000 monthly interactions.

Hoopfer Lab, Carleton College

June 5 - August 25

Machine Learning Research Assistant

Northfield, Minnesota

- Improved pose tracking and behavior analysis accuracy by 30% through the creation of skeletal models with SLEAP deep learning software, advancing Drosophila research outcomes.
- Streamlined data workflows using TensorFlow, NumPy, and Matplotlib, reducing data preprocessing and model training time by 25% for more efficient and accurate analysis.

Virtual Viking Longship Project

July 2023 - April 2024

Junior Software Developer

Northfield, Minnesota

- Developed interactive Viking Age VR simulations for enhanced historical education, increasing engagement in classroom demos by 50%.
- Designed a Unity-based rowing mini-game and implemented mixed reality elements, enabling immersive learning experiences that received positive feedback from 95% of users in testing.

Projects

Carls Chat

- Led a team of 4 to create a campus chat platform featuring real-time messaging, user registration, and campus mapping, achieving 1,500+ student interactions in the first month.
- Engineered a secure and scalable backend using WebSocket, Docker, and Flask to support reliable, real-time campus-wide interactions.

House Price Prediction Model

- Built a Shiny app to predict California house prices, aiding users in data exploration and decision-making; implemented Random Forest and spatial joins to achieve 90% accuracy in predictions.
- Created an intuitive interface with ggplot2 and plotly, improving user engagement with housing data insights by 45% through enhanced visualization and accessibility.

HeadStarter AI Software Engineering Fellowship Projects

- **Pantry Tracker App:** Developed a smart inventory tracker using Google Gemini API, helping users reduce grocery waste by 20%.
- **AI Flashcards:** Designed customized study tools, integrating Stripe for seamless payments and Clerk for secure logins, supporting 500+ active users weekly.
- **AI ChatBot:** Built an interactive AI ChatBot in React, achieving a 4.8/5 user satisfaction rating and supporting over 300 user conversations weekly.
- **Rate My Professor AI:** Created a professor evaluation platform, with 85% of users reporting improved decision-making on professor selection.

Student Organizations

Girls Who Code Loop at Carleton College *Founder and President*

Gender Minorities in Mathematics and Statistics *Board Member*

Skills

Languages : Python, Java, HTML5/CSS, R, MATLAB, Kotlin, Ruby

Databases : PostgreSQL, SQLite, Firebase, MongoDB

Libraries : React, NumPy, TensorFlow, Pandas, PyTorch, Matplotlib, NLTK, Tidyverse

Frameworks : Flask, Bootstrap, Django

Tools : Git, Node.js, Docker